

understanding of what customers really wanted. They brainstormed new ideas for product experiments that might have more impact. In fact, many of these ideas were not new. They had simply been overlooked because the company was focused on building social tools. As a result, Grockit tested an intensive solo-studying mode, complete with quests and gamelike levels, so that students could have the choice of studying by themselves or with others. Just as in Farb's original classroom, this proved extremely effective. Without the discipline of split testing, the company might not have had this realization. In fact, over time, through dozens of tests, it became clear that the key to student engagement was to offer them a combination of social and solo features. Students preferred having a choice of how to study.

Kanban

Following the lean manufacturing principle of kanban, or capacity constraint, Grockit changed the product prioritization process. Under the new system, user stories were not considered complete until they led to validated learning. Thus, stories could be cataloged as being in one of four states of development: in the product backlog, actively being built, done (feature complete from a technical point of view), or in the process of being validated. Validated was defined as “knowing whether the story was a good idea to have been done in the first place.” This validation usually would come in the form of a split test showing a change in customer behavior but also might include customer interviews or surveys.

The kanban rule permitted only so many stories in each of the four states. As stories flow from one state to the other, the buckets fill up. Once a bucket becomes full, it cannot accept more stories. Only when a story has been validated can it be removed from the kanban board. If the validation fails and it turns out the story is a bad idea, the relevant feature is removed from the product (see the chart on [this page](#)).